

NAVYASRI AKULA

Data Science / AI Engineer

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PROFESSIONAL SUMMARY

Data Science graduate with hands-on experience across machine learning, deep learning, computer vision, and NLP, delivered through end-to-end analytics and AI projects. Skilled in Python, SQL, and Power BI, with experience building predictive models, image classification pipelines, and information-retrieval systems. Trained in Generative AI and LLM fundamentals (prompt engineering, RAG) and currently expanding into Agentic AI (LangChain, LangGraph, CrewAI). Seeking Data Science / AI Engineer roles to apply strong ML fundamentals and a fast, self-directed learning record.

TECHNICAL SKILLS

Languages: Python, SQL

Machine Learning & Deep Learning: Scikit-learn, TensorFlow, Supervised Learning, Feature Engineering, Model Evaluation, Explainable AI (Grad-CAM)

NLP, GenAI & Agentic AI: Text Preprocessing, Semantic Similarity, Query Expansion, LLM Fundamentals, Prompt Engineering, RAG, Agentic AI (in progress)

Data Analysis & Visualization: Pandas, NumPy, Power BI, Tableau, Matplotlib, Seaborn, Exploratory Data Analysis (EDA)

Tools & Platforms: Git, GitHub, Jupyter Notebook, Google Colab, OpenCV, Streamlit

PROFESSIONAL EXPERIENCE

Data Science Intern

Jan 2024 – May 2024

- Diagnosed key drivers of employee attrition by analyzing job-satisfaction and performance data, then engineered supervised ML models to flag at-risk employees for HR retention teams.
- Cleaned and preprocessed HR datasets — handling missing values and encoding categorical variables — to boost model input quality and prediction reliability.
- Built and deployed interactive Power BI dashboards tracking attrition trends across departments and roles, giving HR stakeholders faster visibility into turnover risk.
- Designed credit and debit card analytics dashboards to monitor customer spending behavior, transaction trends, and financial KPIs.
- Analyzed the BigMart sales dataset (8,500+ records) to identify product and outlet factors driving revenue performance.

PROJECTS

AI-Driven Customer Intelligence & Revenue Analytics System

Python, Machine Learning, Power BI

- Engineered an end-to-end analytics pipeline processing multi-source transactional data to surface revenue insights.
- Segmented customers into 4 RFM-based groups (Champions, Loyal, At Risk, Hibernating) to prioritize retention strategy.
- Built Customer Lifetime Value (CLV) models to forecast future revenue contribution per customer.
- Developed a churn prediction model to flag high-risk customers, enabling proactive retention outreach.
- Delivered an interactive Power BI dashboard tracking revenue trends, growth rate, and customer segments.

Apple Leaf Disease Detection & Severity Analysis

TensorFlow, OpenCV, MobileNetV2, Streamlit

- Engineered a MobileNetV2-based deep learning model achieving 99.1% test accuracy classifying apple leaf diseases from images.
- Applied image preprocessing and computer vision techniques to boost detection accuracy across multiple disease classes.
- Used Grad-CAM to visualize the leaf regions driving each prediction, then estimated disease severity via infected-area calculation and deployed the model through a Streamlit app; collaborated with a 3-person team.

Adaptive Query Expansion System (ASQE)

Python, NLP, TF-IDF, Word2Vec

- Designed an adaptive query expansion framework using TF-IDF, Word2Vec, and pseudo-relevance feedback to improve search relevance on the AG News dataset (120K+ articles across 4 categories).
- Applied semantic similarity methods to dynamically enrich and refine user queries based on context.
- Improved retrieval quality and relevance of search results over baseline keyword matching.

EDUCATION

Integrated B.Sc–M.Sc in Data Science

2021 – 2026

VIT-AP University — CGPA: 8.62

Intermediate

2019 – 2021

Govt. Junior College for Girls — CGPA: 8.53

10th (SSC)

2018 – 2019

Modern High School — CGPA: 8.8

CERTIFICATIONS & PUBLICATIONS

- Research Paper: EcoLens (DistilBERT-LightGBM) — NCRTAD 2025
- Deloitte & Quantum Data Analytics Job Simulations — Forage
- Python for Data Science — Great Learning